

considering solutions to the heat equation with a space-time periodic, zero-mean in space source $f(t, x)$,

$$\partial_t u - \Delta u = f. \quad (1.6)$$

The long time behavior of the solution to this equations is described by what is called the pullback attractor, which, in this case, is a single trajectory

$$u^{\mathcal{A}}(t, x) := \int_{-\infty}^t e^{(t-\tau)\Delta} f(\tau, x) d\tau = \int_{t_0}^t e^{(t-\tau)\Delta} f(\tau, x) d\tau,$$

for some t_0 , which we assume is zero appropriately translating f in time. Here the second equality follows from the time periodicity of $u^{\mathcal{A}}(x, t)$. All the trajectories (solutions to (1.6)) converge to $u^{\mathcal{A}}(t, x)$ in the pullback sense (as the initial time goes to minus infinity). This holds even for the 3D NSE with small forces, see [11].

If spatial oscillations of f dominate the temporal oscillations, then $(-\Delta)^{-1}f(t, x)$ is the leading term in $u^{\mathcal{A}}(t, x)$. However, in applications to convex integration, the non-diagonal interactions between building blocks result in errors of type

$$f = P_H \operatorname{div} (w \otimes w),$$

where w is the velocity (or magnetic field) perturbation, which, for simplicity, is assumed to be a just a T -periodic in time building block. To control the size of the error Δw , the perturbation has to be highly intermittent (with the intermittency dimension $D < 1$). Then it is easy to check that

$$\|(-\Delta)^{-1} \operatorname{div} (w \otimes w)\|_{L_x^2} \gg \|w\|_{L_x^2}.$$

Thus, temporal oscillations of the perturbation w have to dominate spatial ones.

Let

$$\bar{f}(x) := \frac{1}{T} \int_0^T f(\tau, x) d\tau, \quad \partial_t^{-1} f(t, x) := \int_0^t (f(\tau, x) - \bar{f}(x)) d\tau,$$

where T is the time period of f . Note that $\partial_t^{-1} f(t, x)$ is also time periodic. Then

$$u^{\mathcal{A}}(t, x) = \int_0^t e^{(t-\tau)\Delta} \bar{f}(x) d\tau + \partial_t^{-1} f(t, x) + \int_0^t e^{(t-\tau)\Delta} \Delta \partial_t^{-1} f(\tau, x) d\tau. \quad (1.7)$$

When the frequency of temporal oscillations is high enough, $(-\Delta)^{-1} \bar{f}(x)$, or the first term in (1.7) is the leading term of $u^{\mathcal{A}}(t, x)$. Since time averaging of f traveling along geodesics results in a larger intermittency dimension $D > 1$, we get

$$\|u^{\mathcal{A}}\|_{L_x^2} \ll \|w\|_{L_x^2},$$

and hence $u^{\mathcal{A}}$ can be used as a temporal corrector to cancel the error f .

Notations. Let $\mathbb{N}_+$ denote the set of positive integers. For $p \in [1, \infty]$ and $s \in \mathbb{R}$, we set

$$L_x^p := L^p(\mathbb{T}^3), \quad W_x^{s,p} := W^{s,p}(\mathbb{T}^3), \quad H_x^s := H^s(\mathbb{T}^3),$$

where $W_x^{s,p}$ is the Sobolev space and $H_x^s = W_x^{s,2}$. We also use L_σ^2 for divergence free functions in L_x^2 . Given any Banach space $\mathbb{X}$, we denote by $C([0, T]; \mathbb{X})$ the space of continuous functions from $[0, T]$ to $\mathbb{X}$, equipped with the norm $\|u\|_{C_T \mathbb{X}} := \sup_{s \in [0, T]} \|u(s)\|_{\mathbb{X}}$. In particular, for $N \in \mathbb{N}_+$ we set

$$\|u\|_{C_{T,x}^N} := \sum_{0 \leq m+|\zeta| \leq N} \|\partial_t^m \nabla^\zeta u\|_{L^\infty([0, T]; L_x^\infty)},$$

where $\zeta = (\zeta_1, \zeta_2, \zeta_3)$ denotes the multi-index and $\nabla^\zeta := \partial_{x_1}^{\zeta_1} \partial_{x_2}^{\zeta_2} \partial_{x_3}^{\zeta_3}$. We also use the product spaces

$$\mathcal{L}_x^p := L^p(\mathbb{T}^3) \times L^p(\mathbb{T}^3), \quad \mathcal{H}_x^s := H_x^s \times H_x^s, \quad \mathcal{C}_x^1 := C_x^1 \times C_x^1.$$

Throughout this paper the notation $a \lesssim b$ means that $a \leq Cb$ for some constant $C > 0$.